

Total No. of Questions : 8]

SEAT No. :

PC2799

[6352]-23

[Total No. of Pages :2

S.E. (Electronics /E & TC) /(Electronics and Computer)

PRINCIPLES OF COMMUNICATION SYSTEMS

(2019 Pattern) (Semester- IV) (204193)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Differentiate Between NBFM and WBFM . [6]
b) Draw and explain the block diagram of super heterodyne FM receiver.[6]
c) FM wave is represented by following equation, [6]

$$V = 10 \sin [10^8 t + 3 \sin 10^4 t] \text{ calculate}$$

- i) Carrier and modulating frequency
- ii) Modulation index and frequency deviation
- iii) Power dissipated by FM wave in 100Ω resistor

OR

- Q2)** a) Explain with suitable diagram importance of pre-emphasis and de-emphasis. [6]
b) What are methods of FM generation? Explain any one in detail. [6]
c) An angle modulated signal is described by the equation [6]
 $V = 10 \cos (2\pi F_c t + 4 \sin 2\pi F_m t)$
 $F_c = 10 \text{ MHz}, \quad F_m = 1,000 \text{ Hz}$
 - i) Determine Modulation index and transmitted signal bandwidth.
 - ii) If F_m is doubled, Determine Modulation index and transmitted signal bandwidth.

- Q3)** a) Explain types of sampling with waveform. [6]
b) Explain with block diagram and waveforms generation of PAM. [6]
c) Find the Nyquist rate and Nyquist interval for the following signal [5]
 $X(t) = 3 \cos (200\pi t) + 5 \sin (6000 \pi t) + 10 \cos (1200 \pi t)$

OR

- Q4)** a) Compare PAM, PWM and PPM [6]
b) What is aliasing and Draw a spectrum showing aliasing and guard band.[6]
c) Explain Principles of time division multiplexing? Why synchronization is needed in TDM system. [5]

P.T.O.

- Q5)** a) Draw and explain Block diagram of DM Transmitter. [6]
 b) What are different types of Quantization? Explain any one method of quantization with neat diagram. [6]
 c) What is necessity of companding? Explain A law companding in detail. [6]

OR

- Q6)** a) Draw and explain Block diagram of Adaptive delta modulation. [6]
 b) Compare PCM, DM and ADM. [6]
 c) Explain need of digital communication. [6]

- Q7)** a) Draw the following data formats for bit stream 10110100101 [5]
 i) Unipolar RZ
 ii) Unipolar NRZ
 iii) polar RZ
 iv) Polar NRZ
 v) split phase Manchester
 b) What is equalizer? Explain Adaptive Equalizer. [6]
 c) What is inter symbol Interference (ISI)? Explain methods to eliminate it. [6]

OR

- Q8)** a) Draw and explain CCITT hierarchy multiplexing system. [6]
 b) Explain need of synchronizer in digital multiplexing. Explain frame synchronization. [6]
 c) The data stream [1 1 1 1 1] is given to scrambler shown below. Determine the output sequence of scrambler. Assume initial contents of register to be zero. [5]

